

Control work 4 — percentages, and work on the mistakes

The three questions, the multiplier and the reverse case, under time.

LESSON 74–75

2 × 40 MIN

MATEMATIKA 6, NAZORAT ISHI 4

S7 10 REVIEW

LESSON CLOCK

3

35'

12'

25'

5'

Instructions	3 min
The paper	35 min
Answers	12 min
Diagnosis and rewrite	25 min
The map	5 min

LEARNING OBJECTIVES

- Convert between percentages, fractions and decimals.
- Find a part, a whole and a percentage.
- Handle an increase, a decrease and a reverse question.
- Classify each lost mark and rewrite the whole solution.

TEXTBOOK

National	pp. 154–210
Cambridge	Stage 7, pp. 96–104
Workbook	Control paper 4

01

Explanation

The paper — 25 marks, 35 minutes

Q	TASK	MARKS	FROM
1	Write 35% as a fraction in lowest terms and as a decimal	3	L55
2	Find 18% of 350	3	L59–62
3	63 is 18% of a number: find it	4	L59–62
4	Write 42 as a percentage of 56	3	L59–62
5	Increase 80 000 by 15%	3	L63–66
6	After a fall of 20% a price is 64 000: find the original	4	L63–66
7	A trader buys at 40 000 and sells at 50 000: find the profit percentage	5	L67–70

THE ANSWERS

$\frac{7}{20}$ and 0.35; 63; 350; 75%; 92 000; 80 000; 25% profit.

Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
fraction not simplified	$\frac{35}{100}$	$\frac{7}{20}$
decimal point misplaced	3.5	0.35
multiplied instead of divided	$63 \cdot 0.18$	$63 \div 0.18$
part and whole swapped	$\frac{56}{42}$	$\frac{42}{56}$
increase found but not added	12 000	92 000
reverse question added back	$64\,000 + 12\,800$	$64\,000 \div 0.8$
profit on the selling price	$\frac{10\,000}{50\,000} = 20\%$	$\frac{10\,000}{40\,000} = 25\%$

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

The chapter as one map

QUESTION	OPERATION	SENSE CHECK
a part	multiply	smaller than the whole
the whole	divide	larger than the part
the percentage	$\frac{\text{part}}{\text{whole}} \cdot 100$	under 100 if it really is a part
increase	$\cdot (1 + \frac{p}{100})$	larger
decrease	$\cdot (1 - \frac{p}{100})$	smaller
reverse	divide by the multiplier	multiply forwards to check

LOOKING FORWARD

Percentages return in the pie chart in Quarter IV, where 360° is shared out, and in every later problem about money. The next two lessons are the decimals that all this arithmetic needs.

02 Worked examples

EXAMPLE 1 Model answer, Q3: 63 is 18% of a number.

① The whole is wanted, so divide.

② $63 \div 0.18$

③ $= 350$

④ Check: $0.18 \cdot 350 = 63 \checkmark$

ANSWER

350

EXAMPLE 2 Model answer, Q6: after –20% a price is 64 000.

① The multiplier was 0.8.

② $64\,000 \div 0.8$

Divide, never add back.

③ $= 80\,000$

Check: $80\,000 \cdot 0.8 = 64\,000 \checkmark$

ANSWER

80 000

EXAMPLE 3 Model answer, Q7: bought at 40 000, sold at 50 000.

- ① Profit: 10 000.
- ② The base is the cost price.
- ③ $\frac{10\,000}{40\,000} \cdot 100$
- ④ = 25%

ANSWER

25% profit

03 Interactive model

Return Q6 and Q7 first; both are decided by which number is the base, and naming the base is what the class needs to practise.

The chapter in eight questions

35% as a fraction in lowest terms:

$$\frac{35}{100}$$

$$\frac{7}{20}$$

$$\frac{1}{3}$$

$$\frac{3}{5}$$

35% as a decimal:

3.5

0.35

0.035

35

18% of 350:

63

68

54

35

63 is 18% of:

11.34

350

81

413

42 as a percentage of 56:

75%

133%

42%

24%

80 000 increased by 15%:

92 000

12 000

80 015

68 000

After -20% it is 64 000; before:

76 800

80 000

51 200

84 000

Bought 40 000, sold 50 000:

20%

25%

10%

125%

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	U'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Percentage	Foiz	Процент
Multiplier	Ko'paytuvchi	Множитель
Whole	Butun	Целое
Increase	Ortish	Увеличение
Original value	Boshlang'ich qiymat	Первоначальное значение
Profit	Foyda	Прибыль
Diagnosis	Tashxis	Диагностика

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Q1 needs the fraction:

as written

in lowest terms

as a decimal only

as a percentage

Q3 is:

multiplication

division

addition

a percentage

Q4 puts underneath:

42

56

100

14

Q5 ends with:

the increase

the new amount

the percentage

the multiplier

Q6 is:

multiplication

division

subtraction

addition

Q7 divides by:

the selling price

the cost price

the profit

100

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. 35% as a fraction

ANSWER $\frac{7}{20}$

2. 35% as a decimal

ANSWER 0.35

3. 18% of 350

ANSWER 63

4. 63 is 18% of

ANSWER 350

5. 42 as a percentage of 56

ANSWER 75%

6. 80 000 up 15%

ANSWER 92 000

7. After -20% it is 64 000: before

ANSWER 80 000

Medium · the standard question

7 PROBLEMS

1. Bought 40 000, sold 50 000

ANSWER 25% profit

2. 12% of 2 500

ANSWER 300

3. 96 is 40% of

ANSWER 240

4. 27 as a percentage of 45

ANSWER 60%

5. 600 down 35%

ANSWER 390

6. After +25% it is 500: before

ANSWER 400

7. Bought 60 000, sold 54 000

ANSWER 10% loss

Hard · stretch and reason

7 PROBLEMS

1. Up 20% then down 25%

ANSWER -10%

2. A price after 40% off is 66 000: the original

ANSWER 110 000

3. 84 as a percentage of 70

ANSWER 120%

4. A salary rises 8% to 2 700 000: the old salary

ANSWER 2 500 000

5. 2.5% of 16 000

ANSWER 400

6. Which is worth more: 15% of 80 or 80% of 15?

ANSWER Equal — both 12

7. Why is the base always named first?

ANSWER Every percentage is a percentage of something

07 Homework

Homework — 5 tasks

Rewrite every question you lost a mark on in full, from the first line.

1. Rewrite in full every question on which you lost a mark.
2. Write 64% as a fraction in lowest terms and as a decimal.
3. Find 24% of 750, and say what 180 is as a percentage of 750.
4. After a rise of 30% a price is 91 000. Find the original.
5. A trader buys at 120 000 and sells at 138 000. Find the profit percentage.